Data bases 2

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Data bases 2, 2025/2026

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  - Dipartimento di Elettronica Informazione e Bioingegneria (building 20), 1st floor

- Teaching assistants:
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  - Francesco Invernici francesco.invernici@polimi.it

Data bases 2, 2025/2026

Data	Dove	09:00	10:00	11:00	12:00
Lunedì	26.1.5			DATA BASES 2 (dal *15/09/2025* al *22/12/2025*)	
Martedì					
Mercoledì					
Giovedì					
Venerdì	26.1.2			DATA BASES 2 (dal *19/09/2025* al *19/12/2025*)	
Sabato					

- Lectures and exercise sessions may take place on both days
- According to School's recommendations:
  - No streaming / attendance is only in presence
    → streaming on the days of the public transport strike
  - Recording of (possibly old) lessons is provided

Textbooks in Italian

[BD1 + DB2] P. Atzeni, S. Ceri, P. Fraternali, S. Paraboschi, R. Torlone

"Basi di dati" (2023)

EBOOK available at VitalSource Bookshelf (https://resolver.vitalsource.com/)

From outside Polimi check how to access it: https://www.biblio.polimi.it/en/services/how-to/access-to-an-electronic-resource

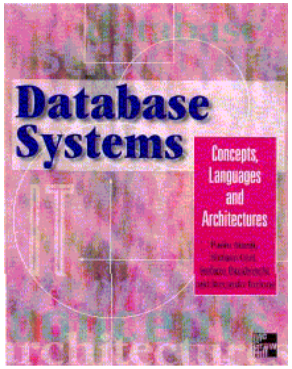
[DB2] P. Atzeni, S. Ceri, P. Fraternali, S. Paraboschi, R. Torlone

"Basi di dati: Architetture e linee di evoluzione" (2003)

Textbooks in English

[DB1 + DB2] P. Atzeni, S. Ceri, S. Paraboschi, R. Torlone

"Database systems" - McGraw-Hill (1999)



DOWNLOADABLE: http://dbbook.dia.uniroma3.it/

Compared to the last Italian version some chapters are missing

More bibliobraphy on specific topics (e.g., JPA) listed in the official program page of the course

Teaching material

- Material is available on the WeBeep portal

  https://webeep.polimi.it/

  under “Data bases 2”
- Materials include slides of the lectures, exercise sessions, forum, etc.

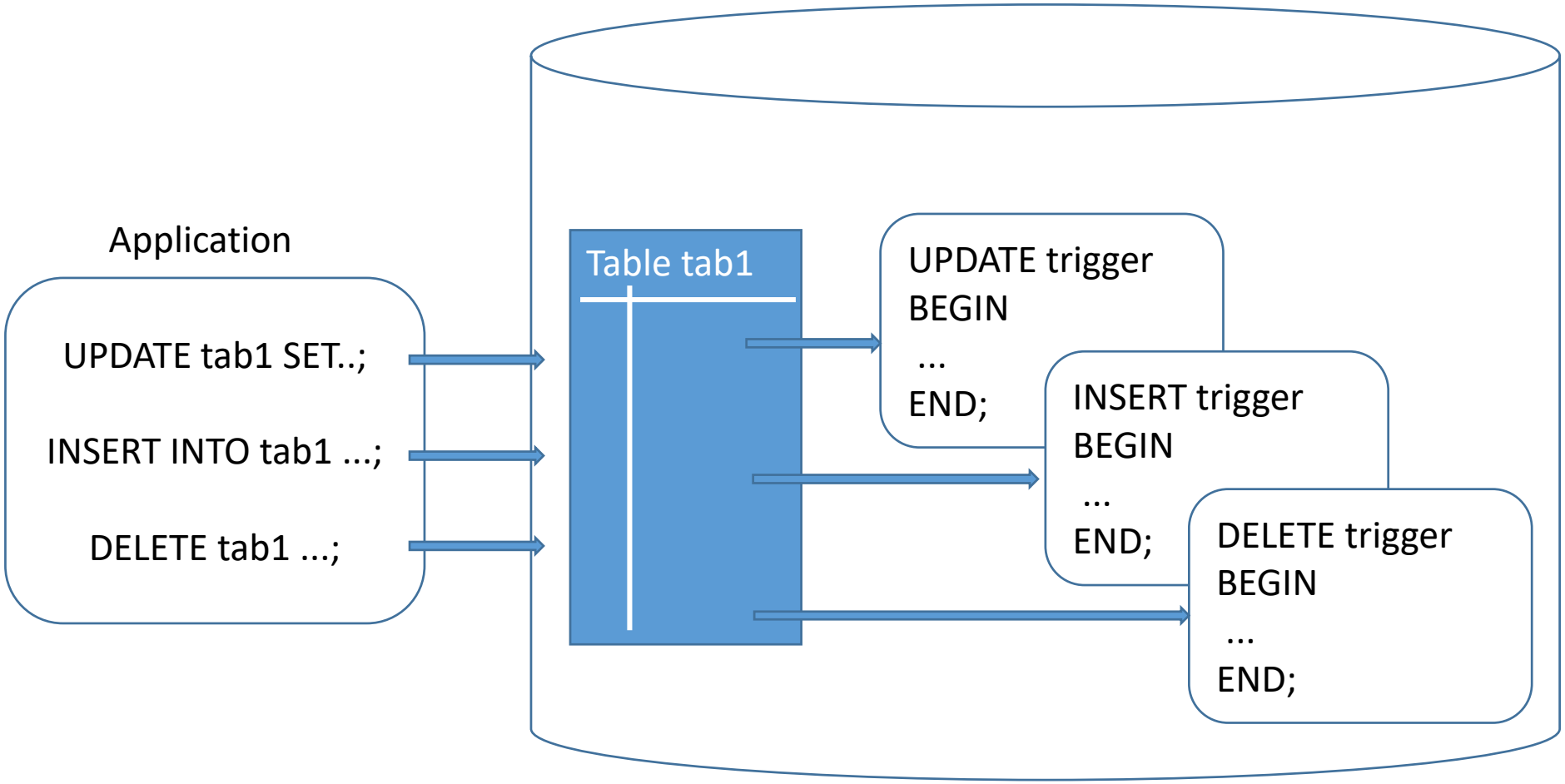
Prerequisites

- Basics of Database systems
  - Relational model
  - SQL (SQL-92)
  - Relational algebra

    → See [DB1 + DB2] textbooks

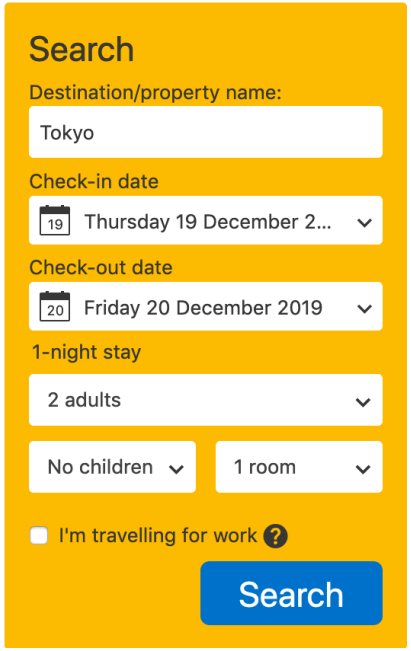
  - Object oriented design with UML and programming with Java (basic)

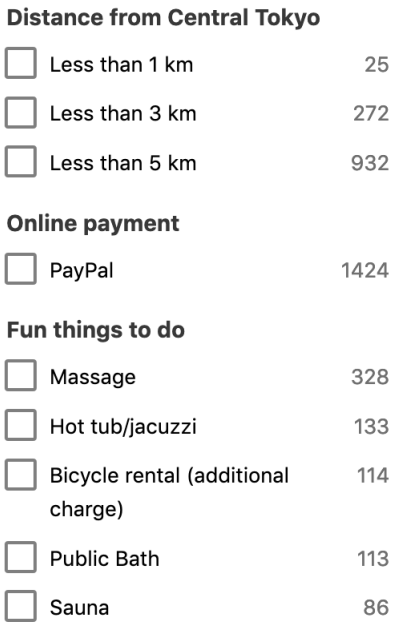
Triggers (SQL)

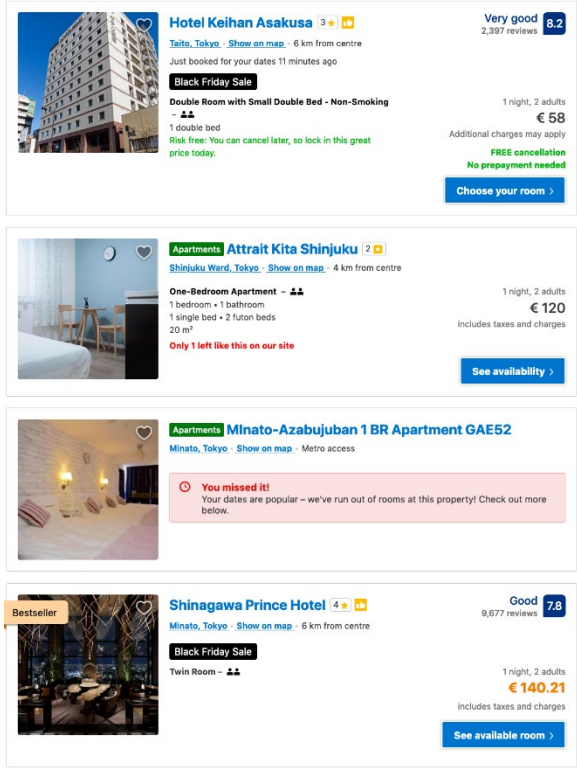


Ranking data

- We are able to store lots of data, but how do we get the *best* data out of a dataset?







- How does SQL accommodate ranking?

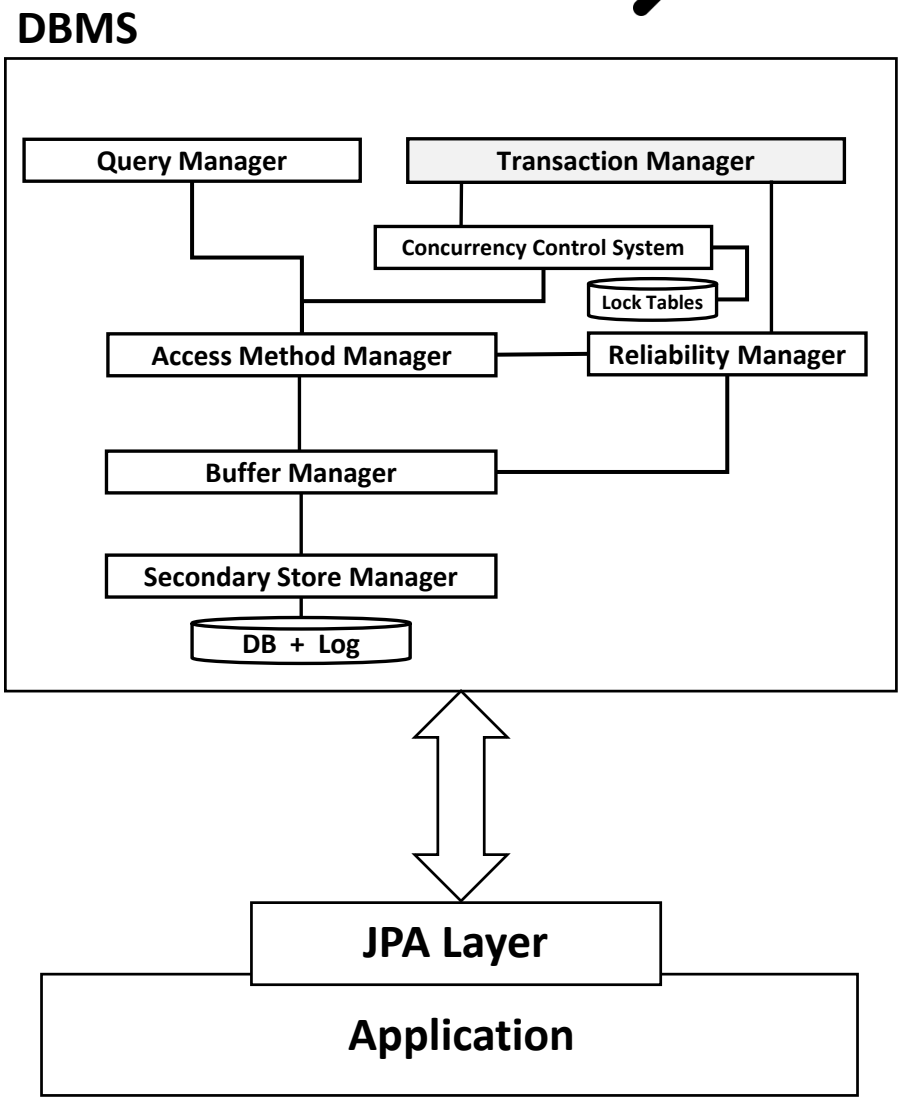
Database architecture and ORM

Opening the database box

- Physical data management and query optimization
- Transaction management
  - Concurrency control
    - CC theory, locking techniques
  - Reliability control
    - Log and recovery

Opening the application box

- Mapping objects to tables and methods to transactions
- Managing the alignment between object state in main memory and data state in secondary memory
- Building advanced queries



Related topics addressed in other courses

- **056901 - SYSTEMS AND METHODS FOR BIG AND UNSTRUCTURED DATA**
  - Principles and approaches for data storage and management
  - Unstructured data management (NoSQL)
    - graph, semantic, column, document, key-value, IR- based databases
  - Design methodologies for the specification of NoSQL data models

- **056895 - STREAMING DATA ANALYTICS**
  - From data streams and time-series to complex event recognition and processing
  - Streaming Data analytics techniques

Exam

- The exam consists of a **written closed-book** verification covering all the topics of the course (<u>everything published on WeBeep</u>, including materials for flipped classrooms)
  - **3 exercises** on the 5 main topics, possibly with related theoretical questions
    1. Triggers
    2. Ranking and skylines
    3. Physical DBs/Query optimization
    4. DB architecture/technologies (concurrency control etc.)
    5. ORM and JPA

- **No oral exams**

Exam rules

- During the exam:
  - No books, notes, or electronic devices are allowed
  - A calculator is allowed
  - Cheating policies: Communication with other students is strictly prohibited. In this case, students will be asked to exit the classroom (both!).

- In the event of plagiarism, students are notified to the appropriate offices and are postponed to a next exam *session*

- Positive marks:

  If a mark is rejected and the written exam is retaken (simply sitting down and seeing the exam text is sufficient), the previous mark is lost!

Exam rules

Please, check the general rules for the exams at Politecnico:

https://www.polimi.it/en/students/didactic-activities-and-degree-examination/exams

Exam – mobility students

- Mobility students may take the DB2 exam at the foreign university during their mobility period, at the same time as the exam in Milan, with the **invigilation of an official staff member** (professor or other permanent staff at university)
- Prerequisite: finding such a local officer and putting me in contact with him/her **at least one week before the exam**
  - No exceptions will be allowed

Tentative agenda

Date	Time	#ore	Lec/Exe	Topic	Teacher
15/09/2024	10:15-12:15	2	Lecture	Triggers	SC
19/09/2024	10:15-12:15	2	Lecture	Triggers	SC
22/09/2024	10:15-12:15	2	Exercises	Triggers	FI
26/09/2024	10:15-12:15	2	Lecture	Physical DB	SC
29/09/2024	10:15-12:15	2	Exercises	Triggers	FI
03/10/2024	10:15-12:15	2	Lecture	Physical DB	SC
06/10/2024	10:15-12:15	2	Lecture	Physical DB	SC
10/10/2024	10:15-12:15	2	Exercises	Physical DB	DB
13/10/2024	10:15-12:15	2	Exercises	Physical DB	DB
17/10/2024	10:15-12:15	2	Lecture	Concurrency Control	SC
20/10/2024	10:15-12:15	2	Lecture	Concurrency Control	SC
24/10/2024	10:15-12:15	2	Lecture	Concurrency Control	SC
27/10/2024	10:15-12:15	2	Exercises	Concurrency Control	DB
31/10/2024	10:15-12:15	2	Lecture	Concurrency Control	SC
03/11/2024	10:15-12:15	2	Exercises	Concurrency Control	DB
07/11/2024	10:15-12:15		Mid-term (other) exams		
10/11/2024	10:15-12:15		Mid-term (other) exams		
14/11/2024	10:15-12:15	2	Lecture	Ranking	SC
17/11/2024	10:15-12:15	2	Exercises	Concurrency Control	DB
21/11/2024	10:15-12:15	2	Lecture	Ranking	SC
24/11/2024	10:15-12:15	2	Exercises	Ranking	FI
28/11/2024	10:15-12:15	2	Lecture	JPA/ORM	TBD
01/12/2024	10:15-12:15	2	Lecture	JPA	SC
05/12/2024	10:15-12:15	2	Exercises	Ranking/JPA	FI
08/12/2024	10:15-12:15		Holiday		
12/12/2024	10:15-12:15	2	Exercises	JPA	FI
15/12/2024	10:15-12:15	2	Exercises	Q&A - exercises	SC
19/12/2024	10:15-12:15	2	Lecture	Reliability (flipped)	
22/12/2024	10:15-12:15				

Check WeBeep
for the updated version

+10h tutoring